

Boise State University
Department of Electrical and Computer Engineering
ECE 350L Signals and Systems Lab
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Lab Report

Experiment #4: Matched Filters and Sonar for Localization

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1. INTRODUCTION

The goal of this report is to understand the theory of correlation receivers, and how they are implemented as matched filters. The matched filter that is designed in this report is used for an active sonar application. The accuracy of the matched filter is analyzed with varying standard deviations of the Additive White Gaussian Noise (AWGN).

2. BACKGROUND THEORY AND PROCEDURE

The goal of a receiver is to accurately decipher what signal was transmitted, even with the presence of noise. While there are many solutions for recovering corrupted signals, a correlation receiver is used for our active sonar application.

The correlation receiver is the product of the expected signal and the corrupted signal. In Figure 1.a, the expected signal (the dashed line) is aligned with the corrupted signal (the solid line). Figure 1.b is the product of the correlation receiver, outputting a graph that is positive. When the integration of Figure 1.b is taken, a large positive number is outputted, which means the two signals are aligned and we have received our expected signal.

In the other case, Figure 2.a shows us two misaligned signals. Figure 2.b examines the product of the two signals; it's found that the line contains negative values. The integration of this figure will be a negative, telling us the two signals are not aligned, and we have not received our signal.

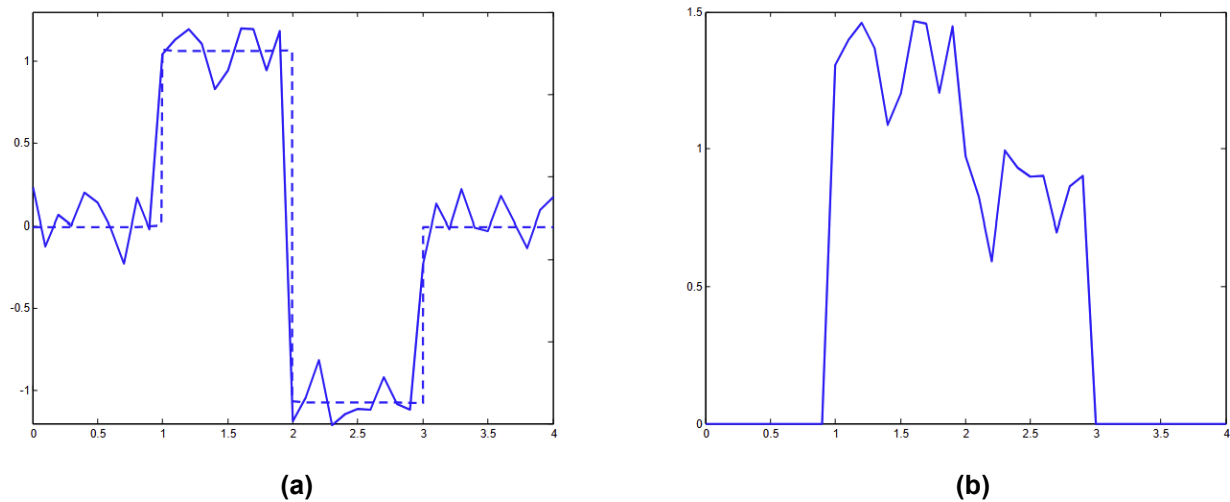


Figure 1: (a) Expected signal is aligned with corrupted signal. (b) The product of the expected signal and the corrupted signal when aligned.

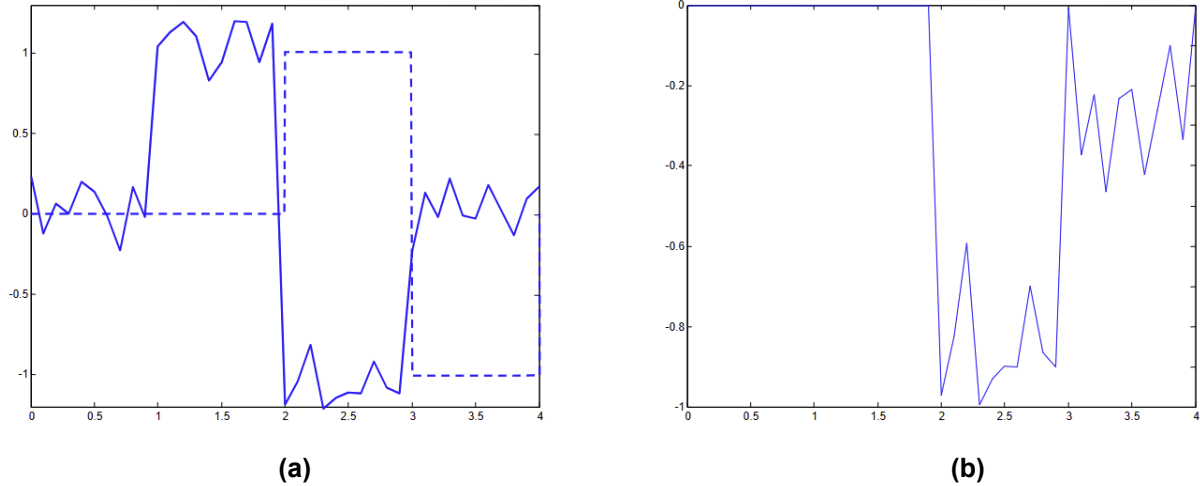


Figure 2: (a) Expected signal misaligned with the corrupted signal. (b) The product of the expected signal and the corrupted signal when misaligned.

Correlation receivers are difficult to implement in the analog domain. Instead, a matched filter will be used, whose impulse response is the time-reversed version of the expected signal. The matched filter output is mathematically the same as the correlation

$$c(t)|_{t=T} = \int_0^T x(\tau)s_1(\tau)d\tau \quad (1),$$

but the filter implementation is more practical.

The active sonar will use this matched filter, which operates by sending a “ping” of sound and listening for return echoes from objects. This will be simulated in MATLAB to test the accuracy of the matched filter with different standard deviations of AWGN.

Figure 3 shows a sinusoid wave that is 0.25 seconds long. From Figures 1.a and 1.b, this sinusoid will be our “expected signal”, or as $s_1(\tau)$ in equation 1. The second signal, $x(\tau)$, refers to the received signal which will be our original signal with added white Gaussian noise, at standard deviations of 0.5, 1, and 2.

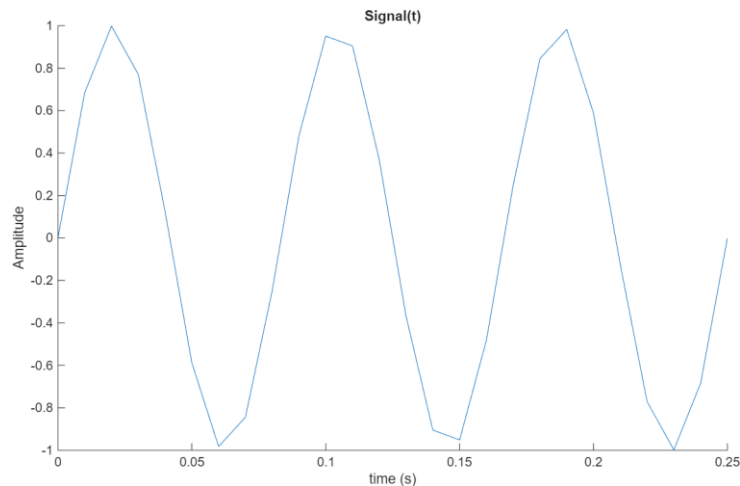


Figure 3: Expected Signal, Sinusoid with Length of 0.25 seconds

For our specific use of this lab, we will use the transmitted signal to find distances underwater. We will do this by transmitting the signal and receiving the signal that has reflected off the object. Figure 4 is an example of what may be received from the receiver, using our sinusoid from Figure 3. In this example, we can use the equation

$$distance = \frac{t(s)}{2} * 1500 \left(\frac{m}{s} \right) \quad (2)$$

to find the distance of objects. In this equation, t is the total time it took to receive the ping, and 1500 m/s comes from the assumption that sound travels at 1500 m/s underwater. We divide time by 2 because when the signal reaches the object, it must return to the receiver.

To find when we have received the signal more reliably, especially with the presence of AWGN, we will need to implement the matched filter. In Figure 5, we have the convolution of the signal and the received data from Figure 4.

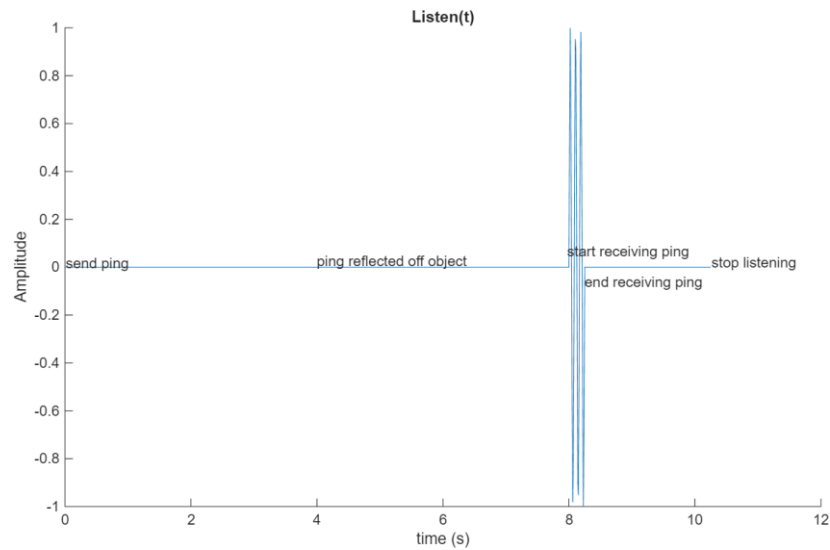


Figure 4: Plot of the Received Data from the Receiver

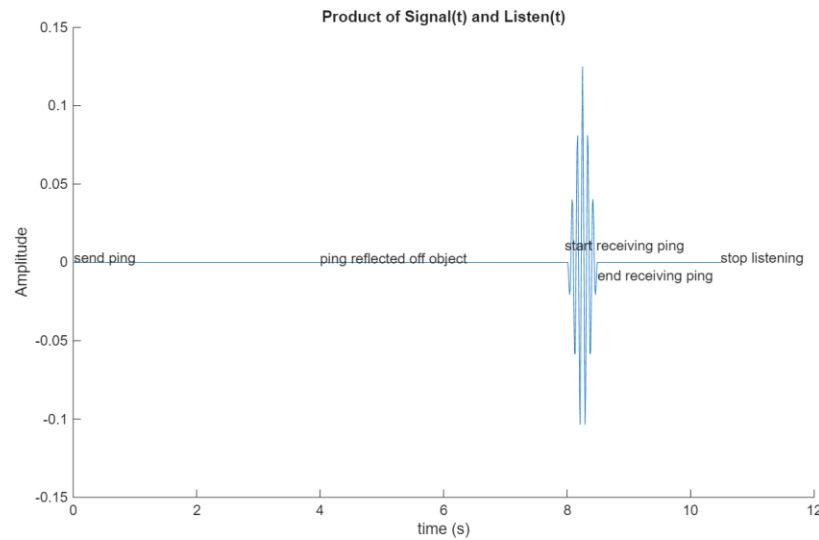


Figure 5: Convolution of Expected Signal and Received Data

It is important for receivers to be able to distinguish the transmitted signal from the noise. A matched filter is used in the application of active sonar, with the goal of finding distances underwater. It is then tested and simulated with varying standard deviations of AWGN.

3. DATA ANALYSIS AND INTERPRETATION

In the Data Analysis and Interpretation section, the accuracy of the matched filter will be analyzed when deciphering the expected signal from different standard deviations of AWGN. The signal we want to extract is the same from Figure 4, which should give us 6000 meters. The standard deviations used will be 0.5, 1, and 2, with ten total trials for each.

3.1 Received Data with Noise

Signals in the real world are often disrupted by noise, which can come from natural sounds such as marine-life or underwater earthquakes in our case. In our simulation, we will assume Additive White Gaussian Noise and will add the noise to what the receiver is listening to.

In Figure 6, there are 3 graphs with varying standard deviations of 0.5, 1, and 2. In each of the plots, our received signal will stay the same as shown in Figure 4. In figure 6.a, we can somewhat pinpoint the signal we want to extract around the 8 second marker.

The difference in standard deviations of AWGN does change what we find on our different plots. As standard deviations of the AWGN increase, the amplitudes of the plots are more extreme and more prone to bigger swings in change. In figure 6.a, the standard deviation is set to 0.5 and we obtain peak amplitudes of around 1. In figure 6.b, the standard deviation increases to 1, and we obtain bigger peak amplitudes of around 3.

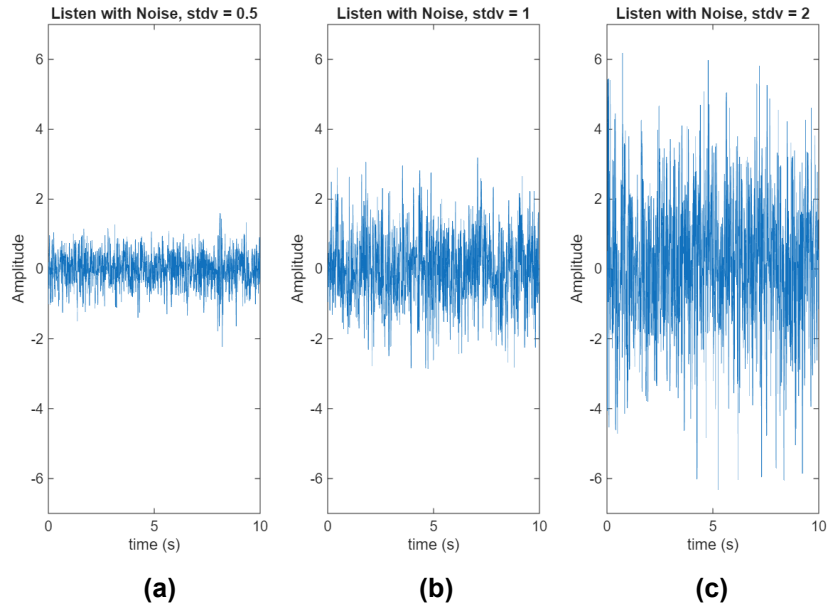


Figure 6: (a) Received data with AWGN, $\sigma = 0.5$. (b) Received data with AWGN, $\sigma = 1$. (c) Received data with AWGN, $\sigma = 2$.

In the real world, noise plays a crucial role in signal processing. It is important for engineers to understand the different types of noise and how they interact with certain systems. In our case, we are analyzing our system with varying degrees of AWGN.

3.2 Distances Calculated

The goal of our matched filter is to correctly pinpoint our expected signals with the presence of noise. Using the same standard deviation values of AWGN from section 3.1, ten trials will be run on each to test for accuracy of the system.

The value we expect is 6000 meters; if our result matches that, the trial was successful. In Table 1, the table is organized vertically by standard deviation of the AWGN and is organized horizontally by trial. The success rate with standard deviation of 0.5 is 100%, the success rate with standard deviation of 1 is 50%, and the success rate with a standard deviation of 2 is 0%.

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Trial 7	Trial 8	Trial 9	Trial 10
$\sigma=0.5$	6000.0	6000.0	6000.0	6000.0	6000.0	6000.0	6000.0	6000.0	6000.0	6000.0
$\sigma=1$	6000.0	5992.5	6000.0	495.0	6000.0	6060.0	6000.0	6000.0	2017.5	4080.0
$\sigma=2$	5422.5	5992.5	1747.5	4447.5	1725.0	7447.5	5880.0	2977.5	5227.5	232.5

Table 1: Distances Calculated through Match Filtering

From the results of Table 1, there is a negative linear correlation between the success rate of the matched filter and the standard deviation of the AWGN. In other words, as we increase the standard deviation of the noise, our matched filter has a lower chance of successfully finding our expected signal.

It is important for engineers to understand how certain systems are affected by noise. In our results, the matched filter struggled to correctly pinpoint the signal as standard deviation of noise increases.

4. CONCLUSION

In this report, we examined the theoretical foundations of correlation receivers and demonstrated how these principles enable the design and implementation of matched filters. By applying these concepts to an active sonar system, we were able to evaluate how effectively the matched filter detects signals embedded in AWGN. In the analysis, varying standard deviations of noise to better understand how the system is affected by AWGN.